Part I. Random and Stochastic ODE

Sec1. Introduction.

- Einstien 1900s.
- Langevin. $\frac{dx}{dt} = f(x) + \sigma(x)\eta_t$.

$$E(\eta_t) = 0 \qquad E(\eta_{t_1}\eta_{t_2}) = \rho\,\delta(t_1 - t_2)$$

- Itô: stochastic ODE.

Sec 1. - 1). simple numeric scheme
2). Taylor expansion. — ODE, SODE, RODE. $\Rightarrow$ Sec 6.

Sec 2. - Existence of RODE.

$$\frac{dx}{dt} = f(x, \eta_t(\omega))$$

3 Thm.

- 1) η_t : continuous $\Rightarrow$ 1. Picard-Lindelöf, 2. Peano's — $x(t)$.
- 2) η_t : measurable. $\Rightarrow$ Carathéodory sense.

3. $x(t)$: absol cont.

1.1. Simple numeric scheme for RODE.

$$\frac{dx}{dt} = -x + \eta_t.$$

- Euler scheme.

$$\int_{t_n}^{t_{n+1}} \Rightarrow \quad x(t_{n+1}) - x(t_n) = \int_{t_n}^{t_{n+1}} (-x(s) + \eta_s)\, ds$$

$$\underbrace{x(t_{n+1})}_{x_{n+1}} - x(t_n) = -x(t_n) + \eta_{t_n}$$

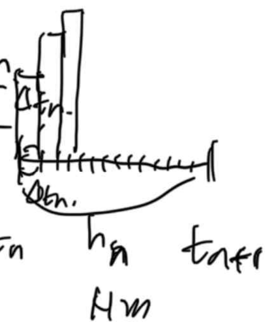
$$h_n := t_{n+1} - t_n$$

$$x_{n+1} = (1-h_n)x_n + h_n \eta_{t_n}$$

$$\Downarrow \quad \int_{t_n}^{t_{n+1}} \eta_s\, ds$$

- Averaged Euler scheme.

$$x_{n+1} = (1-h_n)x_n + \sum_{j=1}^{N_{h_n}} \eta_{t_n + j\Delta t_n} \Delta t_n$$



- $\dfrac{dx}{dt} = f(x) + \sigma(x)\eta_t$. — SODE

$$x_{n+1} = x_n + h_n f(x_n) + \sigma(x_n) \int_{t_n}^{t_{n+1}} \eta_s\, ds$$

$$= x_n + (f(x_n) + \sigma(x_n) I_n) h_n.$$

$$I_n = \frac{1}{h_n} \int_{t_n}^{t_{n+1}} \eta_s\, ds$$

- $\dfrac{dx}{dt} = q(x, \eta_t)$

$$x_{n+1} = x_n + \int_{t_n}^{t_{n+1}} q(x(s), \eta_s(\omega))\, ds \quad \text{implicit.}$$

$$x_{n+1} = x_n + q(x_n, I_n) h_n.$$

- Taylor expansion

1.2 Taylor expansion

- To derive higher order scheme
- Check the order of convergence of scheme.

1.2.1 ODE. $\dfrac{dx}{dt} = f(t,x)$

Total derivative $u(t,x)$ $x(t)$

$$Du(t,x) := \frac{du}{dt}(t,x) + f(t,x)\frac{du}{dx}(t,x)$$

f_x x' x'' $\vdots$

Df f_t $\dfrac{dx}{dt} = f$. $x^{(p)}$

t, x

$$x^{(j)}(t) = \frac{d^j x}{dt^j} = D^{j-1} f(t, x(t))$$

$x(t)$ $f(t,x)$ continuint

$\dfrac{dx}{dt} = f(x, x(t))$ - conti.

where $D^0 = Id$. $x(t)$ $x'(t) =$

Let f: $p+1$ times conti-diff., $\exists\, \tau \in [t_0, t_0+h]$.

$$x(t_0+h) = x(t_0) + x'(t_0)h + \cdots + \frac{x^{(p)}(t_0)}{p!}h^p + \frac{x^{(p+1)}(\tau_n)}{(p+1)!}h^{p+1}$$

Def). p-Taylor approximation of the solution value

$x(t_0+h) \simeq x(t_0+h, x_0, h_0)$

$$\Phi(t_0, x_0, h) = x_0 + \sum_{j=1}^{p} \frac{h^j}{j!} D^{j-1} f(t_0, x_0)$$

1.2.2. SODE case

x_t: Holder continuous. $x(t)$:

$$x(t) = x(t_0) + \int_{t_0}^{t} f(s, x(s))\, ds$$

1st taylor

p-th

$$= x_0 + \int_{t_0}^{t} f(t_0, x(t_0))\, ds + \int_{t_0}^{t}\int_{t_0}^{s} Df(\tau, x(\tau))\, d\tau\, ds$$

$\therefore$ $x(t)$: dose not second derivative

$$x(t) = x_0 + \sum_{\sigma=1}^{p} D^{\sigma-1} f(t_0, x_0) \int_{t_0}^{t}\int_{t_0}^{s_1} \cdots \int_{t_0}^{s_{\sigma-1}} ds_\sigma \cdots ds_1$$

$$+ \int_{t_0}^{t}\int_{t_0}^{s_1} \cdots \int_{t_0}^{s_\sigma} D^p f(s_{\sigma+1}, x(s_{\sigma+1}))\, ds_{\sigma+1} \cdots ds_1$$

1.2.3 RODE case $\quad z_t$.

$$\underline{\frac{dx}{dt} = \underline{g}(x, z_t(\omega))}, \quad := \underline{G_\omega(t,x)} \quad ; \text{ may not diff int.}$$

(with Ω above ω)

g : cont ftn. w.r.t both variable

$\underline{t} \to g(x(t), z_t(\omega(t)))$: not diff

$(x, \underline{\omega}) \to \underline{g}(x, \omega)$: diff.

$$\underline{g}(x(s), w(s)) = \sum_{|d| \le p} \underset{\frac{1}{d_1!}\cdot\frac{1}{d_2!}}{\frac{1}{d!}}\, \partial^{d} g(x_0, w_0)\, (\underset{x(s)-x_0}{\Delta x_s})^{d_1} (\underset{w(s)-w_0}{\Delta w_s})^{d_2} + R_{p+1}(s)$$

($\partial_x^{d_1} \partial_w^{d_2}$ above ∂^d)

$$\Delta x_t = x(t) - x_0 = \int_{t_0}^{t} \underline{g(x(s), w(s))}\, ds$$

$$= \int_{t_0}^{t} \sum_{|d| \le p} \frac{1}{d!} \partial^{d} g(x_0, w_0) (\Delta x_s)^{d_1} (\Delta w_s)^{d_2}\, ds$$

$$+ \cancel{\int_{t_0}^{t} R_{p+1}(s)\, ds}$$

$$\int_{t_n}^{t_{n+1}} \Rightarrow x_{n+1} - x_n = \int_{t_n}^{t_{n+1}} \sum_{|\alpha| \le p} \frac{1}{\alpha!} \partial^\alpha q(x_0, w_0) (\Delta x(s))^{\alpha_1} (\Delta w_s)^{\alpha_2} ds$$

when $p = 0$.

$$x_{n+1} - x_n = h_n q(x_n, w_n)$$

when $p \ge 1$. Implicit

$p = 2$.

$p = 3$.

when $p = 1$. $\alpha = (1,0)$ $\alpha = (0,0)$ $\alpha = (0,1)$

$$x_{n+1} - x_n = \int_{t_n}^{t_{n+1}} \sum_{\substack{\alpha=(0,0)\\ \alpha=(1,0)}} \frac{1}{\alpha!} q(x_n, w_n) (\Delta x_s)^{\alpha_1} (\Delta w_s)^{\alpha_2}$$

$$= h_n q(x_n, w_n) + \partial_x q(x_n, w_n) \int_{t_n}^{t_{n+1}} \Delta x_s\, ds$$

$$\boxed{\begin{aligned} \Delta x_s = x(s) - x(t_n) &= \int_{t_n}^{s} q(x(s), w(s))\, ds \\ &= (s - t_n) q(x_n, w_n) \end{aligned}}$$

$$= h_n q(x_n, w_n) + \frac{1}{2} h_n^2 \partial_x q(x_n, w_n) q(x_n, w_n)$$

Sec 2. Random ODE.

$$\frac{dx}{dt} = q(x, \eta_t(w)) := G_w(t, x)$$

$\eta : [0,T] \times \Omega \to \mathbb{R}^m$: stochastic process

with cont. sample path

with conti sample path.

$q : \mathbb{R}^d \times \mathbb{R}^m \to \mathbb{R}^d$: conti

- If η_t continuous in t. → G cont t. ⇒ Classic ODE Thm
- If η_t measurable in t → G meas t. ⇒ Caratheodory sense

2.1.1. Classical assumption

Fix a sample path w.r.t ω.

- $\dfrac{dx}{dt} = G(t, x)$ $x(t_0) = x_0$ (diff eqn)

$\Downarrow$ $\Uparrow$.

- $\left(x(t) = x_0 + \int_{t_0}^{t} G(s, x(s))\, ds\right)$ (int eqn)

Lem If $x(t)$: conti & satisfied (int eqn)

⇒ $x(t)$: sol of (diff eqn) & $x(t)$: cont diff.

Thm 2.1. (Picard-Lindelof)

Let $R = \{(t, x) : t \in [t_0, t_0 + a], |x - x_0| \le b\}$

G unif Lipschitz cont in x cont in t.

$|G(t, x)| \le M$. on R.

⇒ (diff) eqn has unique sol x^* on $[t_0, t_0 + \mu]$

$\mu = \min\{a, \frac{b}{M}\}$ (for fixed sample path)

Proof) (Step 1) define $x_n(t)$

$$x_{n+1}(t) = x_0 + \int_{t_0}^{t} G(s, x_n(s))\, ds$$

with $x_0(t) = x_0$.

$\Rightarrow \begin{cases} x_{n+1}(t) : \text{cont on } [t_0, t_0+\alpha] \ (\because x_n : \text{cont}) \\ |x_{n+1}(t) - x_0| \le b \end{cases}$

$\hookrightarrow \because |x_{n+1}(t) - x_0| \le |t - t_0|\,|G|, \le \alpha M \le b.$

(Step 2)

$$|x_{n+1}(t) - x_n(t)| \le \frac{M k^n (t-t_0)^{n+1}}{(n+1)!}$$

$x_n = \quad x_{n-1}$

when $n = 0$.

$$|x_1(t) - x_0(t)| \le \int_{t_0}^{t} G(s, x_0(s))\, ds \le M(t - t_0).$$

when $n \ge 1$

$$|x_{n+1}(t) - x_n(t)| = \int_{t_0}^{t} G(s, x_n(s)) - G(s, x_{n-1}(s))\, ds$$

$$\le k \int_{t_0}^{t} |x_n(s) - x_{n-1}(s)|\, ds \quad \text{(Lipschitz ineq)}$$

$$\le k \int_{t_0}^{t} \frac{M k^{n-1} (s-t_0)^n}{n!}\, ds \quad \text{(Induction)}$$

$$\le M k^n \cdot \frac{(t-t_0)^{n+1}}{(n+1)!}$$

(step 3).

$$x_{n+1}(t) = x_0 + \sum_{i=0}^{n} [x_{i+1}(t) - x_i(t)]$$

$$|x_m(t) - x_n(t)| \le \sum_{i=m}^{n-1} |x_{i+1}(t) - x_i(t)| \le \sum_{i=n}^{n-1} M K^i \frac{(t-t_0)^{i+1}}{(i+1)} < \varepsilon$$

$\exists N, \le n, m$ $\{x_n\}$: unif Cauchy $\Rightarrow$ unif

$$\lim_{n\to\infty} x_n(t) = x^*(t) \quad \text{unif on } [t_0, t_0+m].$$

$$\lim_{n\to\infty} G(t, x_n(t)) = G(t, x^*(t)) \quad \text{unif} \quad "$$

$\because$ $|G| \le m$, DCT

$$x^*(t) = x_0 + \int_{t_0}^{t} G(s, x^*(s))\, ds$$

(step 4) Uniqueness

Let $Y(t)$ is another sol with $Y(t_0) = x_0$

$$|x_n(t) - Y(t)| \le \frac{M K^n (t-t_0)^{n+1}}{(n+1)!}.$$

$n \to \infty$. $x^*(t) = Y(t)$

Thm 2.2 (Peano's thm)

G : cont in x and t.

$|G(t,x)| \le M$ on R

$\Rightarrow$ (diff eqn) has $\underset{\sim}{\text{at least one}}$ sol

x^* on $[t_0, t_0+\alpha]$ for $\alpha = \min\{a, b/M\}$

Proof) $x_0(t)$, cont diff $[t_0-\delta, t_0]$

- $x_0(t_0) = x_0$
- $|x_0(t) - x_0| \le b$.
- $|x_0(t) - x_0(s)| \le M|t-s|$ for $t, s \in [t_0-\delta, t_0]$

$x_\varepsilon(t)$ for $0 < \varepsilon \le \delta$ on $[t_0-\delta, t_0+\alpha]$

$$x_\varepsilon(t) = \begin{cases} x_0(t) & t \in [t_0-\delta, t_0] \\ \underline{x_0 + \int_{t_0}^t G(s, \underbrace{x_\varepsilon(s-\varepsilon)}_{[t_0-\varepsilon, t_0]})\,ds} & t \in [t_0, t_0+\alpha] \end{cases}$$

$\Downarrow$

continuous

$t \in \underline{[t_0, t_0+\varepsilon]}$, $n\varepsilon = t$

$\Rightarrow [t_0+\varepsilon, t_0+2\varepsilon]$

(Step 2). estimate of $x_\varepsilon(t)$

$$|x_\varepsilon(t) - x_0| = \left|\int_{t_0}^t \underbrace{G(s, x_\varepsilon(s-\varepsilon))}\,ds\right| \le M(t-t_0) \le M\alpha = b.$$

$\Rightarrow \{x_\varepsilon\}$: unif bdd.

$$|x_\varepsilon(t) - x_\varepsilon(s)| = \int_s^t \underbrace{G(\tau, x_\varepsilon(\tau-\varepsilon))}\,d\tau \le M|t-s|$$

$\Rightarrow \{x_\varepsilon\}$: equi cont.

There exist subsequence $\varepsilon_1 > \varepsilon_2 > \dots$ s.t $\varepsilon_n \to 0$ as $n \to \infty$. and

$$\lim_{n\to\infty} x_{\varepsilon_n}(t) = x^*(t) \quad \text{unif on } [t_0-\delta, t_0+\alpha]$$

$$\lim_{n\to\infty} G(t, x_{\varepsilon_n}(t-\varepsilon_n)) = G(t, x^*(t)) \quad //$$

Thus (DCT)

$$x^*(t) = x_0 + \int_{t_0}^{t} G(s, x^*(s))\, ds$$

Chapter 2. Random ODE.

2.1.3. Carathéodory Assumption.

Brief review

Sec 1 Taylor expansion.

- ODE - deri
- SODE - int.
- RODE - $g(x, \eta_t)$,

Sec 2. RODE.

$$\underline{\frac{dx}{dt} = \underline{g(x,\eta_t)} := \underline{G_\omega(t,x)}} \quad \text{(RODE)}$$

with G_ω: ~~cont~~ meas.

wher $\eta : [0,T] \times \Omega \to \mathbb{R}^m$ stochastic process.

$(\Omega, \mathcal{F}, \mathbb{P})$: probability space.

g, G : cont in both variable.

$F(G) \in m.$ $\quad$ $\eta_t.$: cont in $t.$

$\left\{ \begin{array}{ll} \cdot \text{ Picard - Lindelöf} & (G. \text{ Lip in } x) \quad \text{uniq} \\ \cdot \text{ Peano.} & (G. \text{ cont in } t) \quad \cancel{\text{uniq}} \end{array} \right.$

$\boxed{\text{Today}}$

$\cdot$ η_t : cont in $t.$ $\Rightarrow$ g, G_w : cont in t

$\Rightarrow$ $x(t)$ sol (RODE) $\Rightarrow$ x : cont diff.

$\cdot$ η_t : meas in $t.$ $\Rightarrow$ g, G_w : mea in $t.$

absolutely cont

$\Rightarrow$ $x(t)$ sol (RODE) $\Rightarrow$ x : (??)

$\int G_w(t, x)\, dt.$

η_t : meas

Def) $x : [t_0, T] \to \mathbb{R}^d$: absolutely cont.

if for $\forall \varepsilon > 0$, $\exists \delta > 0$ s.t.

$\sum_{i=1}^{n} |x(t_i) - x(s_i)| < \varepsilon$ for $\{[s_i, t_i) : i = 1, \cdots, n\}$

which is finite collection of non-overlapping intervals

in $[t_0, T]$, $\sum_{i=1}^{n} (t_i - s_i) < \delta.$

Note x : absolutely cont $\Rightarrow$ x : weak differentiable almost everywhere

weak deri

$$\int_{t_0}^{T_0} x'(s)\phi(s)\, ds = -\int_{t_0}^{T} x(s)\phi'(s)\, ds$$

$\phi \in C_0^{\infty}((t_0, T), \mathbb{R})$

$\boxed{\text{Lem 2.3}}$ If $f : [t_0, T] \to \mathbb{R}$ bdd, Leb mea, Leb int.

& $x(t) = \int_{t_0}^{t} \boxed{f(s)}\, ds$

$\Rightarrow$ $x(t)$: absolutely cont. & it weak derivate $x'(t) = f(t)$

for almost all $t \in [t_0, T]$.

(Last/. f : cot $\Rightarrow$ $x(t)$ $\Rightarrow$ cont diff)

If x : cont $t \to G(t, x(t))$ Leb int.

$$\underset{\text{absol cont.}}{x(t)} = x_0 + \int_{t_0}^{t} G(s, x(s))\, ds \qquad \text{(IVP)}$$

$$\frac{dx}{dt} = G(t, x(t)), \quad x(t_0) = x_0.$$

Def. A funtion G is said to satisfy Carathéodory Assup

(C1) (cont) $G(t,x)$: cont in x for almost ever $t \in [t_0, T]$.

(C2) (meas) $G(t,x)$: Lebesgue mea. in t. for each $x \in \mathbb{R}^d$

(C3) (bounded) $|G(t,x)| \le m(t)$: absol cont. $t \in [t_0, T]$

Theorem (Carathéodory). Let G satisfy (C) conditions

then the (IVP) has a solution $x^* : [t_0, t_0 + \delta] \to \mathbb{R}^d$.

Proof). $M(t) = \int_{t_0}^{t} m(s)\, ds$; $t \to t_0$, $t \in [t_0, T]$. cont, non-decreasing. $M(t_0) = 0$.

(Step 1) Construct $x_n(t)$. $t \in [t_0 - \delta, T]$.

For $\delta > 0$. If $\delta = 0$ $\delta \to 0$

$t \in [t_0, t_0 + \frac{\delta}{n}]$

x_0,

$$x_n(t) := \begin{cases} x_0 + \int_{t_0}^{t-\frac{\delta}{n}} G(w,s,x_n(s,w))\,ds & t \in [t_0+\frac{\delta}{n}, t_0+\delta] \end{cases}$$

$t \in [t_0 + \frac{\delta}{n}, t_0 + 2\cdot\frac{\delta}{n}]$

$x_1(t) = x_0$

$\Rightarrow t \in [\ \ + \frac{2\delta}{n} \quad + 3\cdot\frac{\delta}{n}]$

$$x_2(t) = \begin{cases} x_0 & [t_0, t_0+\frac{\delta}{2}] \\ x_0 + \int_{t_0}^{t-\frac{\delta}{2}} G(w,s,x_2(s,w))\,ds & [t_0+\frac{\delta}{2}, t_0+\delta] \end{cases}$$

$x_n(t)$: cont function for $t \in [t_0, t_0+\delta]$.

(step 2) Estimate of $x_n(t)$

By (C3)

- $$|x_n(t) - x_0| \le \begin{cases} 0 & \text{if } t \in [t_0, t_0+\frac{\delta}{n}] \\ \int_{t_0}^{t-\frac{\delta}{n}} G(s, x_n(s))\,ds \le \int_{t_0}^{t-\frac{\delta}{n}} n(s) & \text{otherwise} \end{cases}$$

$$\le M(t-\frac{\delta}{n})$$

$$\le \bar{M}.$$

$\therefore$ $x_n(t)$: unif bdd

- $$x_n(t_1) - x_n(t_2) = \begin{cases} 0 & t_1, t_2 \le t_0 + \frac{\delta}{n} \\ \left(x_0 + \int_{t_0}^{t_1-\frac{\delta}{n}} G\,ds\right) - x_0 & t_2 \le t_0 + \frac{\delta}{n} \le t_1 \\ \left(x_0 + \int_{t_0}^{t_1-\frac{\delta}{n}} G\,ds\right) - \left(x_0 + \int_{t_0}^{t_2-\frac{\delta}{n}} G\,ds\right) & t_0 + \frac{\delta}{n} \le t_1, t_2 \end{cases}$$

i) $|x_n(t_1) - x_n(t_2)| \le \left| M(t_1 - \frac{\delta}{n}) - M(t_2 - \frac{\delta}{n}) \right|$

iii) $\quad t_1 - \frac{\delta}{n} \quad\quad \int^{t_1-\frac{\delta}{n}} \quad\quad t_1 > t_2$

$$\int_{t_2-\frac{\delta}{n}}^{t_1-\frac{\delta}{n}} G\,ds \le \int_{t_2-\frac{\delta}{n}} \underbrace{|G|}_{\le m(t)}\,ds \qquad t_1 \ge t_2$$

$$= \left| M\left(t_1-\frac{\delta}{n}\right) - M\left(t_2-\frac{\delta}{n}\right)\right|,$$

ii) $\displaystyle\int_{t_0}^{t_1-\frac{\delta}{n}} G\,ds \le \underline{M\left(t_1-\frac{\delta}{n}\right)} \boxed{-\ M\left(t_2-\frac{\delta}{n}\right)}$,

$t_2-\frac{\delta}{n} \le t_0.$

$$|x_n(t_1) - x_n(t_2)| \le \bar{M}\,|t_1-t_2|.$$

$\therefore$ $x_n(t)$: equicontinuous.

(Step 3) Arzela-Ascoli

$$\lim_{j\to\infty} x_{n_j}(t) = x^*(t) \quad \text{unif on } [t_0, t_0+\delta].$$

By (C1) G : cont in x for almost every fixed t.

$$\lim_{j\to\infty} G(t, x_{n_j}(t)) = G(t, x^*(t)) \qquad \text{a.e.}$$

By (C3) $|G(t, x_{n_j}(t))| \le m(t)$, By DCT

$$\lim_{j\to\infty} \int_{t_0}^{t} G(s, x_{n_j}(s))\,ds = \int_{t_0}^{t} G(s, x(s))\,ds$$

$$\left(x_{n_j}(t) = x_0 + \int_{t_0}^{t} G(s, x_{n_j}(s))\,ds - \underbrace{\int_{t-\frac{\delta}{n_j}}^{t} G(s, x_{n_j}(s))\,ds}_{\to 0,\ |G| \le m(t)}\right.,$$

$$\Rightarrow \quad x^{\#}(t) = x_0 + \int_{t_0}^{t} G(s, x^{\#}(s))\, ds$$

for all $t \in [t_0, t_0 + \delta]$.

Summary.

If η_t: cont ; G cont fn t $\Rightarrow$ $\underline{x(t)}$ cont diff $\left\{ \begin{array}{l} \text{P-L} \\ \text{LP} \end{array} \right.$

If η_t : meas ; G meas in t. $\Rightarrow$ $x(t)$: absol cont

$\Rightarrow$ Carathéodory